MIL-G-15214E(SH)
27 May 1980
SUPERSEDING
MIL-G-15214D(SHIPS)
1 April 1977
(See 6.7)

MILITARY SPECIFICATION

GAGE, DEPTH, WRIST,

DIVER'S, SELF-LUMINOUS DIAL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The wrist depth gages covered by this specification are precision instruments that accurately indicate the depth of their submersion in seawater (specific gravity 1.025) from a free surface to the depth of their respective maximum indications.

1.2 Classification. Diver's wrist depth gages shall be of the following marks and classes, as specified (see 6.2.1):

Mark I - Range of depth indication of 0 to 200 feet (ft) of seawater.
Mark II - Range of depth indication of 0 to 50 ft of seawater.
Class 1 - Meets low magnetic effects requirements of method III of MIL-M-19595.
Class 2 - Without the low magnetic effects requirements of MIL-M-19595.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4220

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-M-19595 - Magnetic Effect Limits for Non-magnetic Equipment Used in the Proximity of Magnetic Influence Ordnance.

STANDARDS

FEDERAL

FED-STD-H28/2 - Unified Thread Form and Thread Series, For Bolts, Screws, Nuts, Tapped Holes and General Applications.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-810 - Environmental Test Methods.

DRAWINGS

MILITARY

NAVAL SEA SYSTEMS COMMAND

NAVSHIPS

608-2482954 - Wrist Depth Gage, Mark II MOD 0 (9 sheets).
810-1844080 - Movement Assembly and General Assembly.
810-1844081 - Diaphragm Assembly and Details.
810-1844082 - Bracket Assembly and Details Bridge Plate Assembly and Details.
810-1844083 - Sector Assembly and Details.
810-1844084 - Casing Assembly and Details, Case Bottom Assembly and Details.
810-1844085 - Dial Assembly and Details.
810-1844086 - Pinion Assembly and Details Indicator Hand Assembly and Details.
810-1844087 - Miscellaneous Parts and Details.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B46.1-1978 - Surface Texture.
Y14.5-1973 - Dimensioning and Tolerancing.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee, Agent, Tariff Publications Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INCORPORATED, AGENT
National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 "P" Street, N.W. Washington, DC 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4 and 6.3).

3.2 Description. Gages shall consist essentially of a pressure sensing element, a movement mechanism, and an indicating dial that are enclosed in an evacuated, pressure proof casing. The gages' function shall not be impaired by vibration, impact, hydrostatic pressure, ambient temperature variations, or underwater explosions, as hereinafter specified.

3.2.1 Mark I gages. Mark I gages shall be in accordance with Drawing 810-1844080 thru 810-1844087, inclusive and shall indicate at a depth of 0 to 200 ft.

3.2.2 Mark II gages. Mark II gages shall be in accordance with Drawing 608-2482954 and shall indicate at a depth of 0 to 50 ft.

3.2.3 Screw thread. Screw threads shall conform to FED-STD-H28/2.

3.2.4 Dimensions and tolerances. Dimensions and tolerances shall be as shown on the applicable drawings (see 3.2.1 and 3.2.2). (ANSI Y14.5-1973 may be used as a guide in interpreting the dimensioning and tolerancing practices used on the referenced drawings.)

3.2.5 Surface roughness, waviness and lay. Surface roughness, waviness and lay requirements shall be in accordance with ANSI B46.1-1978.

3.3 Materials. Materials shall be as specified on the applicable drawings and for class 1 equipment shall also meet the magnetic effect limits specified in method III of MIL-M-19595. Minor variations in the composition of materials specified herein or on the applicable drawing will be acceptable, provided that none of the required properties of the material is affected. Materials shall be free of defects that adversely affect the intended use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

3.3.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 Radiological contamination.

3.4.1 Pointer assembly and dial. Prior to assembly into the gage, neither the pointer assembly nor the dial shall demonstrate the presence of removable radiological contamination in excess of 0.005 becquerel (Bq) of Promethium 147 (Pm147) (see 6.6) when tested in accordance with 4.6.2.

3.4.2 Assembled gage. Subsequent to assembly, with the dial cover removed, the finished gage shall not leak nuclear by-product materials as evidenced by the presence of removable radiological contamination in excess of 0.005 Bq of Pm147 when tested in accordance with 4.6.2.

3.5 Magnetic effects of class 1 gages. The finished class 1 gage shall meet the low magnetic limits specified in method III of MIL-M-19595 (see 4.6.3 and 6.5).

3.6 Bezel guard adhesion. The bezel guard shall be securely bonded to the casing (see 4.6.1.2).

3.7 Brightness. The self-luminous markings on the indicating dial and on the pointer shall be uniformly bright. The initial brightness of dials and pointers shall be 65 (plus or minus 10 percent) microlamberts (uL). When tested as specified in 4.6.4, the brightness of self-luminous markings of assembled gages shall be not less than 46.5 uL.

3.8 Radiation level. The radiation level of the gage shall not exceed 1.00 milliroentgens (mR) per hour at any external surface of the gage when tested in accordance with 4.6.5.

3.9 Vibration. The gage shall withstand the vibration test specified in 4.6.6 without sustaining physical damage and still meet the accuracy requirements of 3.13 and the hydrostatic pressure of 3.12.

3.10 Impact shock. The gage shall withstand the impact shock test specified in 4.6.7 without sustaining physical damage and still meet the accuracy requirement of 3.13 and the hydrostatic pressure of 3.12.

3.11 Dial glass impact. The dial glass shall not chip nor fracture when tested in accordance with 4.6.8.

3.12 Hydrostatic pressure. The gage shall be capable of withstanding the following hydrostatic pressures without signs of distortion, physical damage, or leaks and shall still be capable of conforming to the accuracy requirements of 3.13 (see 4.6.9).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

Mark I - gage pressure of 133.5 ± 1.35 lb/in^2.

Mark II - gage pressure of 66.0 ± 1.00 lb/in^2.

3.13 Accuracy.

3.13.1 Mark I gage. At any instant, at any dial position, and at any ambient temperature between 0 and 32.2 degrees Celsius (^{o}C), the Mark I gage shall indicate the depth of its submergence in seawater with an allowable error which shall be plus or minus 1 ft at zero depth (surface), shall increase in a linear manner with increasing depth to an allowable error of plus or minus 2 ft at 50-ft depth and then shall increase in a linear manner with increasing depth to an allowable error of plus or minus 3 ft at 200-ft depth (see 4.6.10).

3.13.2 Mark II gage. At any instant, at any dial position, at any depth within the range of indication and at any ambient temperature between 0° and 32.2^{o}C, the Mark II gage shall indicate the depth of submergence in seawater with an accuracy of plus or minus 1 ft inclusive of the zero and 50 ft graduations (see 4.6.10).

3.14 Underwater explosion shock. The gage shall be capable of withstanding the underwater explosion shock of 4.6.11, and still meet the accuracy requirements of 3.13.

3.15 Serial number. Unless otherwise specified (see 6.2.1), the serial number shall be a series of Arabic numerals starting in any given year with 0001 and, insofar as is practical, increasing consecutively with the number of gages of that type delivered by the contractor in that given year. Serial numbers of rejected gages may be reused at the option of the manufacturer. The serial number shall be followed by, and shall be considered to include, the year of manufacture and the manufacturer's name or distinctive mark, thus:

0123/1975 John Doe Co.
(number) / (year) (name)

The marking shall not impair the structural integrity of the case.

3.16 Technical data. The contractor shall prepare technical data in accordance with the data ordering documents included in the contract.

3.16.1 Operating and maintenance instruction sheet. An operating and maintenance instruction sheet shall be supplied with each gage. For Mark I gages, refer to Drawing 810-1844080, sheet 2. For Mark II gages refer to Drawing 608-2482954, sheet 9.

3.17 Workmanship. The exterior of the gage shall be smooth and free of projections that will injure human skin or puncture lightweight pure gum sheet rubber. Particular care is required in the fabrication of class 1 gages to assure that materials do not become magnetically contaminated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-108.

4.4 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.4.1 Test reports. First article test reports shall be submitted to the contracting activity (see 6.2.2).

4.4.2 Sample size. Three wrist depth gages of each mark and class shall be subjected to first article inspection.

4.4.3 Inspection routine. The samples shall be subjected to the inspections specified in table I, in the order shown.

TABLE I. First article inspection.

Inspection	Requirement paragraph	Test method paragraph
Visual examination	3.2, 3.3	4.6.1
Bezel guard adhesion	3.6	4.6.1.2
Dimensional examination	3.2.4, 3.2.5	4.6.1.3
Radiological contamination	3.4	4.6.2
Magnetic effect (class 1)	3.5	4.6.3
Brightness	3.7	4.6.4
Radiation level	3.8	4.6.5
Vibration	3.9	4.6.6
Impact shock	3.10	4.6.7

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection. - Continued

Inspection	Requirement paragraph	Test method paragraph
Dial glass impact	3.11	4.6.8
Radiological contamination	3.4	4.6.2
Hydrostatic pressure	3.12	4.6.9
Accuracy	3.13	4.6.10
Underwater explosion	3.14	4.6.11
Accuracy	3.13	4.6.10

4.4.4 Failures. One or more failures shall be cause for refusal to grant first article approval.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A, B, and C inspection.

4.5.1.1 Inspection lot. An inspection lot shall consist of all gages of the same mark and class, presented for inspection at one time.

4.5.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table II.

4.5.1.2.1 Sampling plan. 100 percent of all gages and specified gage parts shall be subjected to Group A inspection.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph
Radiological contamination	3.4	4.6.2
Magnetic effect (class 1 gage)	3.5	4.6.3

4.5.1.2.2 Rejected samples. If any gage or gage part fails Group A inspection, that gage or part shall not be offered for delivery.

4.5.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table III, in the order shown, and the sample shall be selected from inspection lots that have passed group A inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL
Visual examination	3.2, 3.3	4.6.1	1.0
Bezel guard adhesion	3.6	4.6.1.2	1.0
Dimensional examination	3.2.4, 3.2.5	4.6.1.3	0.40
Dial brightness	3.7	4.6.4	0.40
Radiation level	3.8	4.6.5	1/
Accuracy	3.13	4.6.10	0.40

1/ Failure of a sample gage to pass this test shall be cause for submission of 100 percent of the lot to this test. Gages that fail this test shall not be offered for delivery.

4.5.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for inspection level III. The sample size shall be based on the inspection lot size from which the sample was selected from group A inspection. The acceptable quality level (AQL) shall be as specified in table III.

4.5.1.3.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.4 Group C inspection. Group C inspection shall consist of the inspections specified in table IV, in the order shown, and shall be made on sample units which have been subjected to and have passed the groups A and B inspection.

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph
Vibration	3.9	4.6.6
Impact shock	3.10	4.6.7
Dial glass impact	3.11	4.6.8
Radiological contamination	3.4	4.6.2
Hydrostatic pressure	3.12	4.6.9
Accuracy (see 4.5.4.1.2)	3.13	4.6.10

4.5.1.4.1 Sampling plan. For each lot (see 4.5.1.1), the number of gages shown in table V shall be selected at random from among those gages of that lot which have been subjected to, and have passed the groups A and B tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

TABLE V. Sampling for Group C inspection.

Number gages in lot	Number gages in test sample
2 to 8	2
9 to 15	2
16 to 25	2
26 to 40	2
41 to 65	2
66 to 110	3
111 to 180	3
181 to 300	5
301 to 500	5
501 to 800	7
801 to 1,300	10
1,301 to 3,200	15

4.5.1.4.2 Sampling for accuracy test. In addition to the samples of 4.5.1.4.1, either a number of gages equal to not less than 20 percent of the quantity obtained by subtracting the group C samples from the group B samples, or 10 gages, whichever is less, shall also be subjected to the accuracy test of 4.6.10. The acceptance number defective shall be zero.

4.5.2 Packaging inspection. Sample packages and packs and the inspection of preservation packaging, packing and marking for shipment and storage shall be in accordance with Section 5 and the documents specified therein.

4.6 Methods of inspection.

4.6.1 Visual and mechanical inspection.

4.6.1.1 Visual examination. The gage shall be examined for obvious defects that will impair serviceability, such as uneven surfaces, rough edges, chipped or cracked dial glasses, and for compliance with 3.17. Surface texture shall be determined in accordance with ANSI B46.1-1978 (see 3.2.5). Gages with one or more defects shall be declared defective.

4.6.1.2 Bezel guard adhesion. The gage shall be examined to determine adhesion of the bezel guard to the gage case (see 3.6). Gages with guards that show evidence of lack of adhesion to the casing shall be declared defective.

4.6.1.3 Dimensional examination. The gage shall be examined for conformance to the dimensions specified on the applicable drawings (see 3.2.1 and 3.2.2).

4.6.2 Radiological contamination. Each pointer assembly and each dial, prior to their use in gage construction, each test gage as required, and each assembled gage offered for delivery shall be subjected to the following radiological contamination test. All exposed surfaces shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

be throughly wiped with a piece of filter paper of high wet strength and absorption capacity moistened with a 5 percent saline solution. No paper shall be used to wipe more than one test item. This paper shall be allowed to dry. The radioactivity on the paper shall then be measured with an instrument capable of detecting the presence of 0.005 Bq of Pm147, on the paper. Any item, the dried paper for which is found to contain in excess of 0.005 Bq of Pm147, shall have failed the test.

4.6.3 Magnetic effect (class 1 gages). The magnetic effect limits of each class 1 gage shall be determined in accordance with method III of MIL-M-19595.

4.6.4 Brightness. spot brightness of the self-luminous markings of the gage shall be determined photoelectrically. The photoelectric device shall have been calibrated not more than 6 months before the test by a brightness standard that is certified by a Government Laboratory. The gage shall have been kept in complete darkness for at least 8 hours prior to and during the test. Gages that fail to meet the requirements of 3.7 shall be declared defective.

4.6.5 Radiation level. The gage shall be tested to determine conformance to 3.8 by taking eight readings in a plane containing the axis of the cylindrical case of the gage, starting at the center of the dial glass and taking a reading every 45 degrees around the exterior of the gage with a Geiger-Mueller type survey meter using a probe with an end mica window of not less than 0.7 or more than 5.0 milligrams per square centimeter (mg/cm^2) thickness. (Nuclear Chicago Model 2612P and AN/PDR-27 are known to meet these requirements.) The test instrument shall have been calibrated to Cobalt 60 radiation not more than 3 months prior to the test. The end of the probe shall contact the surface of the gage. The reading from the instrument shall be multiplied by 5 to obtain the radiation level. Should any gage demonstrate a radiation level at any point tested that is in excess of the requirements of 3.8, the gage shall have failed the test.

4.6.6 Vibration. The gage shall be vibrated for a period of 5 minutes along each of the three mutually perpendicular axes at an acceleration of 10 gravities (g) and a frequency of 25 to 50 hertz (Hz). The frequency shall be varied from the 25 to 50 Hz, back to 25 Hz in a period of 5 minutes, at the rate of 10 Hz per minute. Gages that show evidence of physical damage or of displacement of the indicator hand to the extent that the accuracy requirements of 3.13 cannot be met shall be declared defective. Gages shall not be wiped off or otherwise cleaned following this test.

4.6.7 Impact shock. The gage shall be subjected to the impact shock test in accordance with method 516, procedure I of MIL-STD-810, except: (a) the impact shock shall be 50g and (b) the impulse time shall be 20 ± 5 milliseconds (ms). Three shocks shall be applied in each of the following positions:

(a) Gage horizontal, dial up.
(b) Gage on edge, dial vertical, mid-range graduation on the vertical axis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

(c) Gage on edge, dial vertical, mid-range graduation on the horizontal axis.

The impact shock shall not damage the gage. Gages that show evidence of physical damage, or of displacement of the indicator hand to the extent that accuracy requirements of 3.13 cannot be met, shall be declared defective. Gages shall not be wiped off or otherwise cleaned following this test.

4.6.8 Dial glass impact. The gage shall be placed flat on a horizontal bed of 1-inch plywood, dial up, with the bezel guard removed. Recesses shall be provided for the wrist strap brackets of the gage to insure even bearing of the case bottom on the horizontal surface. A 5/8-inch-diameter steel ball weighing approximately 0.565 ounce shall be freely dropped from a height of 39 inches: once, to strike the juncture of the dial glass and the bezel; and again, to strike the center of the dial glass. The dial glass shall be examined for evidence of chips or fractures. Gages with chipped or fractured dial glasses shall be declared defective. The bezel guard shall be recemented to the casing subsequent to the radiological contamination test of 4.6.2. Gages shall not be wiped off or otherwise cleaned following this test.

4.6.9 Hydrostatic pressure. Mark I gages shall be subjected to a hydrostatic pressure of 133.5 ± 1.35 lb/in^2g. The Mark II gages shall be subjected to a hydrostatic pressure of 66.0 ± 1.00 lb/in^2g. The gage shall be examined for evidence of distortion, physical damage, leaks, or displacement of the indicator hand to the extent that the accuracy requirements of 3.13 cannot be met. Gages that fail to comply with 3.13 shall be declared defective.

4.6.10 Accuracy. The test shall be conducted at any ambient temperature between 0 and 32.2°C. Gage readings shall be reduced to normal atmospheric pressure, 29.92 inches of mercury. The gages shall be positioned at random attitudes varying from a dial horizontal position to a dial vertical position and subjected to an increasing hydrostatic pressure. The pressure may be simulated by a combination of pneumatic and hydrostatic pressure. The true pressure of seawater (specific gravity 1.025), determined with a mercury manometer or other device of comparable accuracy, shall be recorded at the 0, 10, 25, 50, 100, and 200-ft graduations of the dial gage for Mark I gages and at the 0, 5, 15, 25, and 50-ft graduations of the dial gage for Mark II gages. The pressure shall then be decreased and the true pressure again determined for the same graduations on the dial. This cycle shall be repeated for a total of three times. The arithmetic means of the true pressures of the three cycles (reduced to normal atmospheric pressure) shall be employed to determine conformance with 3.13. Gages that fail to comply with 3.13 shall be declared defective.

4.6.11 Underwater explosion shock test. The explosion shock test shall be conducted as specified hereinafter. In a relatively large, calm, deep body of seawater, the gage shall be submerged to a depth of at least 10 ft suspension from a float. The depth of water under the gage shall be not less than the depth of the water over the gage. Inshore from the gage, at a distance of 225 ft from the gage, a charge of 4.4 pounds of HBX-1, or explosive of equivalent impulse shall be submerged so that the center of the charge is at the same depth as the center of the gage (see

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

figure 1). The charge shall be detonated electrically. The gages shall then be subjected to the accuracy test of 4.6.10.

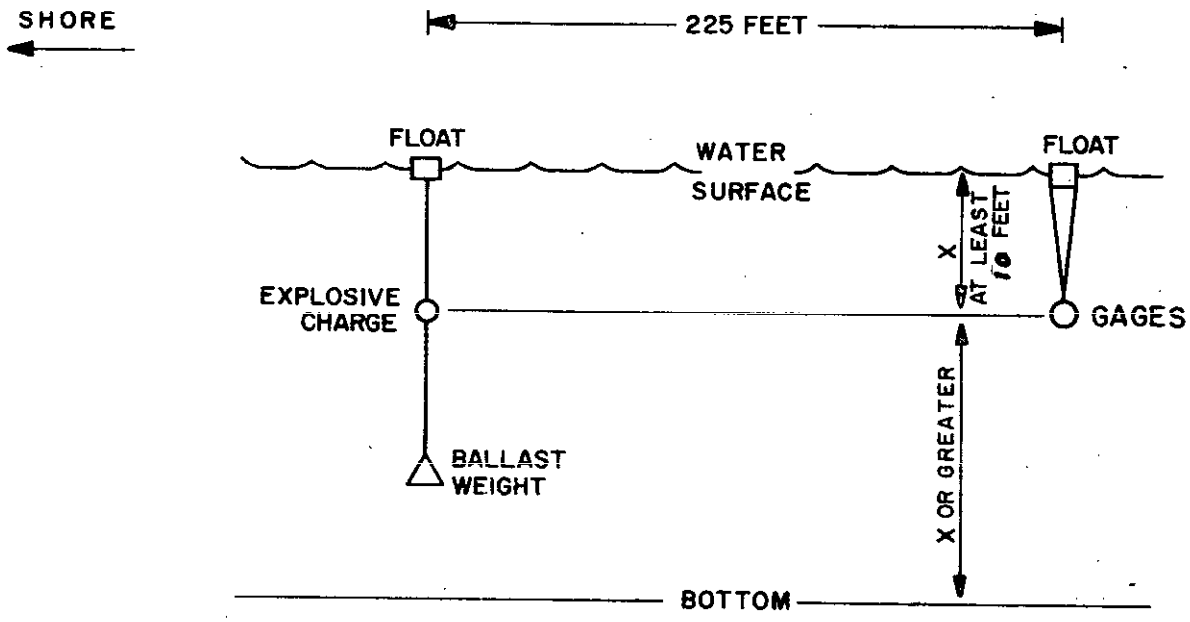


FIGURE 1. Underwater shock blast.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 General requirements. Nonmetallic materials shall be used in accomplishing the method of preservation-packaging and in the construction assembly, closure, waterproofing and reinforcing (packing) for class 1 gages.

5.2 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2.1).

5.2.1 Level A. Each gage, including instruction sheet shall be unit protected in accordance with method III of MIL-P-116. Each gage shall be cushioned, and the gage and instruction sheet shall be placed in a folding or set-up paperboard, fiberboard, or plastic box.

5.2.2 Level C. Each gage, including instruction sheet, shall be packaged to afford protection against deterioration and damage from the supply source to the first receiving activity for immediate use. The contractor's normal preservation-packaging method may be used when such meet the requirements of this level.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.2.1).

5.3.1 Levels A and B. Wrist depth gages, packaged as specified in 5.1 and 5.2, shall be packed in a fiberboard box conforming to class "weather-resistant" boxes of PPP-B-636. Boxes shall be closed, waterproof and reinforced in accordance with the appendix to the box specification with Method V closure applicable. Reinforcing shall be accomplished by the use of nonmetalllic banding or pressure sensitive reinforced tape at the contractor's option. The gross weight shall not exceed the weight limitations of the box specification.

5.3.2 Level C. Wrist depth gages, packaged as specified in 5.1 and 5.2 shall be packed in a container in a manner which will insure safe delivery and acceptance at destination. Shipping containers shall comply with Uniform Freight or National Motor Freight Classification Rules and Regulations or other carrier rules as applicable to the mode of transportation.

5.4 Marking. In addition to any special marking required herein or in the contract (see 6.2.1), interior packages and shipping containers shall be marked in accordance with MIL-STD-129 and MIL-P-116, Table V therein. Marking ink, paint or the material containing ferrous oxide or substances containing any magnetic characteristic shall not be used.

5.4.1 Special marking. Interior packages and shipping containers for class 1 gages shall be marked to include the low magnetic symbol and the words nonmagnetic as follows:

"NONMAGNETIC  (LowMu)"

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

The markings shall be applied in a manner that they are not destroyed when the container is opened.

6. NOTES

6.1 Intended use. The wrist type pressure gages covered by this specification are intended for use by SCUBA divers to indicate the depth of a dive in seawater.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Mark and class of gage (see 1.2).
(c) When a first article sample is required for inspection and approval (see 3.1, 4.4 and 6.3).
(d) If serial number is other than required (see 3.15).
(e) Level of preservation-packaging and level of packing required (see 5.1 and 5.2).
(f) Special marking, if required (see 5.3).

6.2.2 Data requirements. When this specification is used in a contract which invokes the provision of the "Requirements for Data" of the Defense Acquisition Regulation (DAR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the DAR are not invoked in a contract, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data requirements	Applicable DID	Option
3.16.1	Operating and Maintenance	UDI-E-23144	1(Type I, Class 1)
4.4.1	Test reports	DI-T-2072	-----
4.6.4, 4.6.5 and 4.6.10	Calibration of Test Equipment	UDI-T-23742	-----

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 It should be recognized that requirements of low magnetic effects impose stringent material limitations. Materials that provide low magnetic effects meet the requirements of the material specifications specified on the applicable drawing; however, the constituents of alloys may have to be controlled within such specification's limits in order for the finished item to meet low magnetic effects requirements. For example, the magnetic effects of copper alloys depend on their ferritic content to the extent that an item fabricated from a copper alloy of relatively high ferritic content may not meet low magnetic effects requirements.

6.6 Radioactive isotopes are fission products obtainable from and under control of the Isotope Division, Atomic Energy Commission, Oak Ridge Operations Office, P. O. Box E, Oak Ridge, TN 37830.

6.7 Changes from previous issue. The symbol (*) is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4220-N261)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-G-15214E(SH)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.